APPENDIX F

Many-Body Green's Functions

In this appendix we want to give a brief outline of the theory of many-body Green's functions (GF). These techniques are usually not applied very much to nuclear theory, whereas they are widely used in many other many-body areas of work. The advantages of the GF method are that it is very flexible in constructing many-body approximations, and that the GFs have a direct physical interpretation, representing the propagation of particles or holes in the many-body medium. Also, the question of effective interactions can be systematically investigated in this formalism. Because of the brevity of this outline we urge the reader to consult the current literature and text books for standard formulations [Mi 67, AGD 65, No 64a, Ma 67b, FW 71] and we will thus only go into the details where we feel that our representation deviates from the usual one.

F.1 Single-Particle Green's Function and Dyson's Equation

We want to establish the so-called Dyson equation for the single-particle GF, which is defined by:

$$\begin{aligned} G(1,1') &= (-i)\langle 0|T\{a(1)a^{+}(1')\}|0\rangle \\ &= -i\Theta(t_1-t_{1'})\langle 0|a_1(t_1)a_{1'}^{+}(t_{1'})|0\rangle \\ &\quad + i\Theta(t_{1'}-t_1)\langle 0|a_{1'}^{+}(t_{1'})a_1(t_1)|0\rangle. \end{aligned} \tag{F.1}$$

Here and henceforth we use the convention that numbers in brackets comprise all quantum numbers $\{p\}$ to characterize the single-particle state,

such as momentum, spin, isospin, plus time, that is, $(1,2)\triangleq(p_1t_1;\,p_2t_2)$. Repeated arguments shall be summed or integrated (time) over unless otherwise stated. A number as an index comprises everything besides the time. In (F.1), $|0\rangle$ stands for the exact ground state of the A particle system and T is the time ordering operator (C.49). The single-particle operators

$$a^+(1)\equiv a_1^+(t_1)=e^{iHt_1/\hbar}a_1^+(0)e^{-iHt_1/\hbar} \tag{F.2}$$

develop in time along with the full two-body Hamiltonian

$$H=\sum_1\epsilon_1a_1^+a_1+\frac{1}{4}\sum_{1234}\bar{v}_{1234}\,a_1^+a_2^+a_4a_3;\qquad \epsilon_1=\frac{p_1^2}{2m}. \tag{F.3}$$

The usefulness of the single-particle GF of (F.1) comes from the fact that it concisely contains many directly measurable quantities, such as:

(i) The *single particle density*

$$\rho_{11'}=\langle 0|a_{1'}^+a_1|0\rangle=-i\lim_{t_{1'}-t_1\to+0}G(1,1')=\frac{1}{2\pi i}\oint d\omega\,G_{11'}(\omega), \tag{F.4}$$

where $G_{11'}(\omega)$ is the Fourier transform of $G(1,1')$:

$$G_{11'}(\omega)=\int_{-\infty}^{+\infty}d(t_1-t_{1'})e^{i\omega(t_1-t_{1'})}G(1,1'). \tag{F.5}$$

(ii) The *excitation energies of the $A\pm1$ particle system*: This can be made explicit in using a complete set of states in (F.1):

$$\begin{aligned}G_{11'}(t_1-t_{1'})=&-i\sum_{\nu(A+1)}\Theta(t_1-t_{1'})\langle 0|a_1|\nu\rangle\langle\nu|a_{1'}^+|0\rangle e^{-(i/\hbar)(E_\nu^{A+1}-E_0^A)(t_1-t_{1'})}\\&+i\sum_{\nu(A-1)}\Theta(t_{1'}-t_1)\langle 0|a_{1'}^+|\nu\rangle\langle\nu|a_1|0\rangle e^{-(i/\hbar)(E_0^A-E_\nu^{A-1})(t_1-t_{1'})}\end{aligned} \tag{F.6a}$$

or

$$G_{11'}(\omega)=\hbar\sum_{\nu(A+1)}\frac{\langle 0|a_1|\nu\rangle\langle\nu|a_{1'}^+|0\rangle}{\hbar\omega-\left(E_\nu^{A+1}-E_0^A\right)+i\eta}+\hbar\sum_{\nu(A-1)}\frac{\langle 0|a_{1'}^+|\nu\rangle\langle\nu|a_1|0\rangle}{\hbar\omega-\left(E_0^A-E_\nu^{A-1}\right)-i\eta}, \tag{F.6b}$$

where the $|\nu\rangle$ are exact eigenstates of the Hamiltonian (F.3):

$$H|\nu,A\pm1\rangle=E_\nu^{A\pm1}|\nu,A\pm1\rangle.$$

(iii) The *ground state energy*

$$\begin{aligned}E_0=\langle 0|H|0\rangle&=-\frac{i}{2}\lim_{t_1'-t_1\to+0}\sum_1\left(\epsilon_1+i\hbar\frac{\partial}{\partial t_1}\right)G_{11}(t_1-t_{1'})\\&=\frac{1}{2}\oint\frac{d\omega}{2\pi i}\sum_1(\epsilon_1+\hbar\omega)G_{11}(\omega),\end{aligned} \tag{F.7}$$

where we have used the equation of motion for the operators a, a^+:

$$i\hbar\frac{\partial}{\partial t_1}a(1)=\epsilon_1 a(1)+j(1), \tag{F.8a}$$

$$-i\hbar\frac{\partial}{\partial t_1}a^+(1)=\epsilon_1 a^+(1)+j^+(1), \tag{F.8b}$$

with

$$j(1)=\tfrac{1}{2}\bar{v}_{1234}\,a_2^+(t_1)a_4(t_1)a_3(t_1). \tag{F.9}$$

We now proceed to derive Dyson's equation. With the help of Eqs. (F.8) we can establish the equation of motion for the GF:

$$G^{00^{-1}}(1,2)G(2,1')=\delta(1,1')+G(j(1),1') \tag{F.10}$$

with

$$\delta(1,1')=\delta_{11'}\delta(t_1-t_{1'})\hbar,$$

$$G^{00^{-1}}(1,1')\equiv\left[G^{00^{-1}}\right]_{11'}(t_1,t_{1'})=\delta(1,1')\left(i\hbar\frac{\partial}{\partial t_{1'}}-\epsilon_{1'}\right), \tag{F.11}$$

and

$$G(j(1),1')=-i\langle 0|T\{j(1)a^+(1')\}|0\rangle. \tag{F.12}$$

Applying the equation of motion again, we obtain

$$G(j(1),2)G^{00^{-1}}(2,1')=\bar{v}_{1234'}\langle 0|a_2^+a_4|0\rangle\hbar\delta(t_1-t_{1'})+G(j(1),j(1')), \tag{F.13}$$

where we used the form equivalent to (F.11)

$$G^{00^{-1}}(1,1')=\left(-i\hbar\frac{\overleftarrow{\partial}}{\partial t_1}-\epsilon_1\right)\delta(1,1') \tag{F.14}$$

and $G(j(1), j(1'))$ is given by

$$G(j(1),j(1'))=-i\langle 0|T\{j(1)j^+(1')\}|0\rangle. \tag{F.15}$$

The inverse of $G(1,1')$ is defined by

$$G^{-1}(1,2)\cdot G(2,1')=\delta(1,1') \tag{F.16}$$

and we can therefore write

$$G^{00^{-1}}(1,2)G(2,1')=G^{-1}(1,2)\cdot G(2,3)G^{00^{-1}}(3,4)G(4,1'), \tag{F.17}$$

which yields, together with the adjoint of Eq. (F.10),

$$G(1,2)G^{00^{-1}}(2,1')=\delta(1,1')+G(1,j(1')), \tag{F.18}$$

the relation

$$G^{00^{-1}}(1,2)G(2,1')=\delta(1,1')+G^{-1}(1,2)G(2,j(3))G(3,1'). \tag{F.19a}$$

In the same manner we can derive

$$G(1,2)G^{00^{-1}}(2,1)=\delta(1,1')+G(1,2)G(j(2),3)G^{-1}(3,1'). \quad \text{(F.19b)}$$

Multiplying (F.13) from the right with a single-particle GF and using (F.19a), we obtain:

$$G(j(1),1')=\bar{v}_{1234}\langle 0|a_2^+a_4|0\rangle G_{31'}(t_1,t_{1'})+R(j(1),j(2))G(2,1') \quad \text{(F.20)}$$

with [ES 69]:

$$R(j(1),j(1'))=G(j(1),j(1'))-G(j(1),2)G^{-1}(2,2')G(2',j(1')). \quad \text{(F.21)}$$

Inserting the result (F.20) into (F.10) yields, after Fourier transforming,

$$(\hbar\omega-\epsilon_1)G_{11'}(\omega)=\delta_{11'}+M_{12}(\omega)G_{21'}(\omega) \quad \text{(F.22a)}$$

or

$$G(1,1')=G^{00}(1,1')+G^{00}(1,2)M(2,2')G(2',1') \quad \text{(F.22b)}$$

with

$$M_{11'}(\omega)=M^0_{11'}+M^r_{11'}(\omega), \quad \text{(F.23)}$$

$$M^0_{11'}=\bar{v}_{121'4}\rho_{42}, \quad \text{(F.24)}$$

$$M^r_{11'}(\omega)=R_{j(1)\,j(1')}(\omega). \quad \text{(F.25)}$$

Equation (F.22) is the well-known *Dyson equation* and $M(1,1')$ is the corresponding so-called *mass operator*, which has a static (frequency-independent) part M^0 and a dynamic (frequency-dependent) part M^r. The latter is expressed somewhat unusually, and we shall come to its signification later.

It will be the approximation of $M(1,1')$ which will determine the approximation for $G(1,1')$. Let us first make the most simple approximation and drop M^r completely; this yields

$$(\hbar\omega-\epsilon_1)g_{11'}(\omega)=\delta_{11'}+\Gamma_{12}g_{21'}(\omega), \quad \text{(F.26)}$$

$$\Gamma_{11'}=\bar{v}_{121'4}\oint\frac{d\omega}{2\pi i}\,g_{42}(\omega). \quad \text{(F.27)}$$

This constitutes a closed set of nonlinear equations that are equivalent to the Hartree–Fock equation of Chapter 5. This can be seen most easily by inserting $\omega\cdot g_{11'}$ of (F.26) into (F.7) and observing that the resulting expression for the ground state energy is exactly of the Hartree–Fock form (5.40). Also, in moving towards one of the poles of $g_{11'}$ in (F.26) we recover exactly the Hartree–Fock equation (5.44). In analogy to (F.6a, b), the formal solution of (F.26) is then given by

$$g_{11'}(\omega)=\hbar\left(\frac{1-\rho}{\hbar\omega-h+i\eta}\right)_{11'}+\hbar\left(\frac{\rho}{\omega-h-i\eta}\right)_{11'}, \quad \text{(F.28)}$$

where ρ is the HF density operator and h the HF Hamiltonian (5.37).

Analogously, in time–space we have:

$$g(1,1') = -i(\Theta(t_1 - t_{1'})(1-\rho) - \Theta(t_{1'} - t_1)\rho)_{12}(e^{-(i/\hbar)h(t_1 - t_{1'})})_{21'}. \quad \text{(F.29)}$$

In order to better understand the significance of the dynamic part of the mass operator (F.25) we will first show that the complete mass operator corresponds to the most general energy-dependent *optical model potential* for elastic nucleon–nucleus scattering. To show this we work in the momentum representation and write for the elastic scattering S-matrix in the usual way [ES 69]:

$$S_{p_1 p_1'} = {}^{-}\langle p_1 | p_1' \rangle^{+} \quad \text{(F.30)}$$

with

$$\begin{aligned} |p\rangle^{\pm} &= \lim_{t \to \mp\infty} e^{-i(p^2/2m)t/\hbar} a_p^{+}(t)|0\rangle \\ &= \lim_{t \to \mp\infty} e^{-i(E_p - H)t/\hbar} a_p^{+}|0\rangle; \quad \text{(F.31)} \\ E_p &= \frac{p^2}{2m} + E_0^A. \end{aligned}$$

We can therefore express the S-matrix in terms of the single-particle GF of (F.1):

$$S_{p_1 p_1'} = i \lim_{\substack{t \to +\infty \\ t' \to -\infty}} e^{i(p_1^2/2m)t/\hbar} G_{p_1 p_1'}(t - t') e^{-i(p_1'^2/2m)t'/\hbar}. \quad \text{(F.32)}$$

We now rewrite Eq. (F.22) in the following form.

$$\begin{aligned} G(1,1') &= G^{00}(1,1') + G^{00}(1,2)T(2,2')G^{00}(2',1'), \\ T(1,1') &= M(1,1') + M(1,2)G^{00}(2,2')T(2',1'), \quad \text{(F.33)} \end{aligned}$$

where G^{00} is the inverse of (F.11), viz:

$$\begin{aligned} &G^{00}_{p_1 p_1'}(t_1 - t_{1'}) \\ &= -i\delta_{p_1 p_1'} \exp\left[-\frac{i}{\hbar}\frac{p_1^2}{2m}(t_1 - t_{1'}) \right] \cdot \big(\Theta(t_1 - t_{1'}) \cdot \Theta(p_1^2/2m - \lambda) \\ &\quad - \Theta(t_{1'} - t_1) \cdot \Theta(\lambda - p_1^2/2m)\big), \quad \text{(F.34)} \end{aligned}$$

and λ is the Fermi energy, that is, G^{00} is the single-particle GF for a completely free many-body system. The equivalence of (F.33) and (F.22) can be established by simple iteration. Using (F.34) and (F.33) in performing the time limit we find for (F.32):

$$S_{p_1 p_1'} = \delta_{p_1 p_1'} - 2\pi i T_{p_1 p_1'}(\omega = p_1^2/2m)\delta\big((p_1^2 - p_1'^2)/2m\big). \quad \text{(F.35)}$$

We see, therefore, that the matrix $T(1,1')$ defined by Eq. (F.33) corresponds to the scattering T-matrix of elastic nucleon–nucleus scattering. The second of Eqs. (F.33) then also tells us that $M(1,1')$ corresponds to the

optical model potential, which is generally, of course, energy dependent and nonlocal. Usually the optical potential is derived with the help of Feshbach's projection operator formalism (cf. Sec. 4.3.2) using the complete set of shell model states $1p, 2p-1h, 3p-2h, \ldots$ and projecting out the $2p-1h, 3p-2h, \ldots$ components. Therefore, we see that the meaning of the mass operator is that of an "effective" single-particle potential where, within the dynamic part, all couplings to the higher configurations are hidden.* As for all effective potentials, we can also "unfold" the energy dependence for the mass operator and instead write a matrix equation in a larger space (examples will be given later). The mass operator is generally a complex and non-Hermitian quantity and, as usual, special rules have to be observed for the calculation of eigenvectors and eigenvalues of such operators. We do not want to go into these details, and refer the reader to the literature [BD 71, Appendix C].

As we have seen, the calculation of the one-particle GF depends on the knowledge of the mass operator. In order to be able to obtain some approximation for it we first have to learn how to construct a perturbation series for M.

F.2 Perturbation Theory

Before we go into the details of the perturbation theory, we would like to give a very brief outline of how to represent graphically the different expressions we are going to derive with the GF theory. These graphical representations are very useful for grasping at a glance the physical content of an eventually rather complicated equation. We wish to stress the point, however, that we are going to avoid as far as possible deriving relations using purely graphical arguments. Instead, we shall try to derive formulae analytically, especially in the case of perturbation theory.

The single-particle GF (F.1) is a function of two times $t_1, t_{1'}$ consisting of two parts, one with $t_1 > t_{1'}$ and the other with $t_1 < t_{1'}$. We will represent this function on a horizontal time scale by a straight line going from $t_{1'}$ to t_1:

$$G(1,1') \Rightarrow \underset{t_{1'} \quad t_1}{\longrightarrow} \quad \text{or} \quad \underset{t_1 \quad t_{1'}}{\longleftarrow} . \tag{F.36}$$

The arrow gives the line a direction going from $t_{1'}$ to t_1. Defining the time as increasing from left to right, we can then say that "the *particle* propagates from $t_{1'}$ to t_1" if $t_1 > t_{1'}$ and "the *hole* propagates from t_1 to $t_{1'}$" if $t_{1'} > t_1$. In this context we should recognize that the first part of (F.1) just represents the overlap of wave functions with a particle added to the ground state at time $t_{1'}$ and $t_1 > t_{1'}$. In the second part, a hole has been created at time t_1 and $t_{1'} > t_1$. In the pure HF approximation (F.28), only the first part survives if it is a particle (above the Fermi level) and only the

*The equivalence of the optical potential defined by the Feshbach projection operator formalism and the one given here in a perturbative treatment is a somewhat tricky point which we will not treat here.

second one if it is a hole (below the Fermi level). In general, however, both parts contribute to the perturbation series (see below) of a given order.

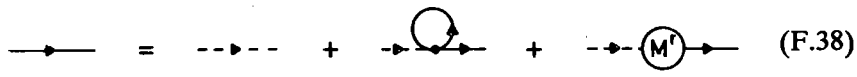
The antisymmetrized matrix elements of the two-body interaction will be represented by a dot:

$$\bar{v}_{1234} \Rightarrow \bullet \tag{F.37}$$

where the dot must always be joined by four lines (one particle GF's) two of them going into the point at 12 and two leaving it at 34.

More particle GF's (see below) will always be represented by as many lines as there are particles (or holes) connected by a bubble.

The Dyson equation (F.22) can now be represented in the following way



(F.38)

with



(F.39)

where R stands for the special combination of GF's figuring in (F.25, 21). The broken line in (F.38) represents the completely free GF G^{00} of Eq. (F.34).

In order to derive perturbation theory analytically (which has the virtue that factors and signs of specific terms can be obtained straightforwardly), we start out from the definition of the so-called multi-time n-body GF:

$$\begin{aligned} &G(1,2,\ldots,n;\,1',2',\ldots n') \\ &= (-i)^n \langle 0|T\{a(1)a(2)\ldots a(n)a^+(n')\ldots a^+(2')a^+(1')\}|0\rangle \\ &= (-i)^n \sum_{\substack{\text{all permu-}\\ \text{tations}}} (-)^P P\{\Theta(t_1-t_2)\cdot\Theta(t_2-t_3)\cdots \\ &\qquad \cdot\Theta(t_{n-1}-t_n)\Theta(t_n-t_{n'})\Theta(t_{n'}-t_{n'-1})\cdots\Theta(t_{2'}-t_{1'}) \\ &\qquad \cdot\langle 0|a(1)a(2)\cdots a(n)a^+(n')\ldots a^+(1')|0\rangle\}. \end{aligned} \tag{F.40}$$

The definition (F.40) implies that the operators are always ordered in such a way that their time arguments increase from right to left. It can also easily be seen that expression (F.40) is completely antisymmetric with respect to the interchange of any two of the indices. We want to write down the equation of motion for the GF, (F.40), which can be easily proved for the one- and two-body case; it is, however, tedious to prove it

in general and we refer the interested reader to the standard literature [Mi 67, AGD 65]:

$$G^{00^{-1}}(1,1'')\cdot G(1'',2\ldots n;1'\ldots n') = \sum_{l=1}^{n}(-)^{l+1}\delta(1,l')\cdot G(2\ldots n;1'\ldots l'-1,l'+1,\ldots,n') - \frac{i}{2}\bar{v}(1,n'+1;1'',n+1)\cdot G\big(1'',2\ldots n,(n+1)^{+};1'\ldots n',(n'+1)^{++}\big). \tag{F.41}$$

A plus sign on an argument means that the corresponding time argument has to be displaced by a positive infinitesimal amount, that is, $1^{+}=p_1,t_1+0$ and $1^{++}=p_1,t_1+0+0$. The matrix element of the two-body interaction is given by

$$\bar{v}(1,2;3,4)=\delta(t_1-t_2)\delta(t_3-t_4)\delta(t_1-t_3)\bar{v}_{1234}. \tag{F.42}$$

It is useful to invert the equation of motion (F.41):

$$G(1\ldots n;1'\ldots n') = \sum_{l=1}^{n}(-)^{l+1}G^{00}(1,l')G(2\ldots n;1'\ldots l'-1,l'+1\ldots n') - \frac{i}{2}G^{00}(1,1''')\bar{v}(1''',n'+1;1'',n+1)\cdot G\big(1'',2\ldots n,(n+1)^{+};1'\ldots n',(n'+1)^{++}\big). \tag{F.43a}$$

The adjoint equation has the form:

$$G(1\ldots n;1'\ldots n') = \sum_{l=1}^{n}(-)^{l+1}G(1\ldots l-1,l+1\ldots n;2'\ldots n')G^{00}(l,1') - \frac{i}{2}G\big(1\ldots n,(n+1)^{++};1'',2'\ldots n',(n'+1)^{+}\big)\bar{v}(n'+1,1'';n+1,1''')G^{00}(1''',1'). \tag{F.43b}$$

We see that the equation of motion relates the n-body GF with the $(n\pm 1)$-body GF's. From Eqs. (F.43) we quite easily derive the expression $G^{00}(1\ldots n;1'\ldots n')$ of the n-body GF in the approximation where we neglect the two-body interaction completely:

$$G^{00}(1\ldots n;1'\ldots n')=\det\big|G^{00}(i,j)\big|. \tag{F.44}$$

It is then clear how the first order expression of an n-body GF has to be calculated: we first calculate the first-order contribution of the one-body GF using (F.44) for the two-body GF. The two-body GF couples to the one-body and three-body GF. The first order of the one-body GF is known from the preceding step, and the three-body GF can be calculated to zeroth order by (F.44); this yields the first order of the two-body GF. We can go on with this scheme to calculate the first order of an n-body GF. To calculate the second order we need the first-order expressions an

we can then set up an analogous scheme. This can then be repeated to any order.

F.3 Skeleton Expansion

For many problems, we are not so much interested in the representation of the expansion terms by $G^{00}(1, 1')$'s, but we want to have an expansion in powers of exact $G(1, 1')$'s (skeleton expansion). That such a partial resummation of the perturbation series is possible can be shown by an investigation of the series involving the $G^{00}(1, 1')$'s only. We can derive the skeleton expansion by multiplying (F.43a) from the left with (F.19b) and using Eqs. (F.9, 12):

$$G(1\ldots n; 1'\ldots n') = \sum_{l=1}^{n} (-)^{l+1} G(1, l') \cdot G(2\ldots n; 1'\ldots l'-1, l'+1\ldots n') - \frac{i}{2} G(1, 1''')\bar{v}(1''', n'+1; 1'', n+1) \cdot R(1'', 2\ldots n, (n+1)^{+}; 1'\ldots n', (n'+1)^{++}), \quad \text{(F.45)}$$

with

$$R(1\ldots n, n+1; 1'\ldots n', n'+1) = G(1\ldots n, n+1; 1'\ldots n', n'+1) - G(1, n+1; 1'', n'+1) G^{-1}(1'', 2'') G(2'', 2\ldots n; 1'\ldots n'). \quad \text{(F.46)}$$

As in the case of the expansion in powers of $G^{00}(1, 1')$, we see from (F.45) that for the skeleton expansion the zeroth order skeleton contribution is given by

$$G^{\text{skeleton}}(1\ldots n; 1'\ldots n') \equiv G^{0}(1\ldots n; 1'\ldots n') = \det|G(i, j)|, \quad \text{(F.47)}$$

that is, to lowest-order skeleton the n-body GF is just the antisymmetrized product of corresponding one-body GFs. Proceeding now in essentially the same way as in the completely free case, we obtain from (F.45, 46, 47) the successive order of the skeleton expansion. We should note thereby that the lowest order of (F.46) is *not* the antisymmetrized product anymore, but the second term on the rhs of (F.46) takes out of the first term just what has already been resumed in the lowest order of the first term on the rhs of (F.45). The reader is invited to convince himself of this fact in taking a definite example.

The above prescription to derive the skeleton expansion is straightforward, yielding the right signs and prefactors of a given order term automatically. Depending on the example and the order, the derivation might be quite tedious. Practice is unavoidable to perform these calculations economically, which is a common feature of all perturbation expansion techniques. In order to become more familiarized with the procedures we propose to verify the following fact: Expression (F.25) for the dynamic

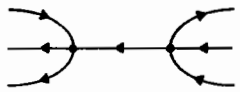


Figure F.1. Second-order contribution to both terms of (F.48).

part of the mass operator contains a generalized three-body GF [ES 69]:

$$R(1,2,3;1',2',3') = G(1,2,3;1',2',3') - G(1,2;1'',2')G^{-1}(1'',1''')G(1''',3;1',3'). \quad \text{(F.48)}$$

Verify that both the first and second terms on the rhs of (F.48) contain (to second order) a term with the structure shown in Fig. F.1 and that they exactly cancel one another. One says that the dynamic part of the mass operator is *one-line irreducible*, that is, it contains no graphs which can be cut into two parts by just cutting one fermion line. A closer investigation shows that *all* graphs of the second term of (F.48) are one-line reducible and these terms cancel equal terms contained in the first term of (F.48). What remains is irreducible.

The second-order skeleton expansion of the mass operator can be easily calculated from (F.48), with (F.47), to give

$$M^{(2)}(1,1') = \tfrac{1}{2}\bar{v}(1,2';1''',2)G(1''',1'')G(2,3')G(3,2')\bar{v}(3',1'';3,1'), \quad \text{(F.49)}$$

which is represented graphically in Fig. F.2. In Fig. F.3 we present very schematically some higher-order terms contained in (F.48). In the drawing we do not present the exchange graphs.

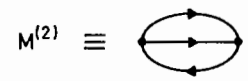


Figure F.2. Graphical representation of the second-order skeleton contribution to the mass operator.

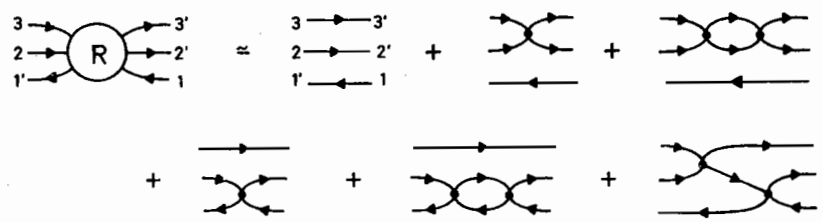


Figure F.3. Higher-order terms contributing to the GF contained in the mass operator.

F.4 Factorization and Brückner–Hartree–Fock

An important feature of n-body GF's is the fact that they approximately factorize into antisymmetric products of lower GF's. This can be verified by an extensive study of the corresponding perturbation series. In Fig. F.3

we see, for example, that the five first terms are just the beginning of the following factorization.

$$R(1,2,3;1',2',3') \simeq G(1,1')\cdot G(2,3;2',3') + \{[G(2,2')\cdot R(1,3;1'3') - (3 \leftrightarrow 2)] - [3' \leftrightarrow 2']\} + \{[G(2,2')\cdot G(1,1')\cdot G(3,3') - (3 \leftrightarrow 2)] - [3' \leftrightarrow 2']\}, \tag{F.50}$$

with

$$R(1,3;1',3') = G(1,3;1'3') + G(1,3')\cdot G(3,1'). \tag{F.51}$$

The function (F.51) is essentially identical with the linear response function defined in (8.130)*; the fact that we have to add the second term in (F.51) can be verified in comparing (F.51) with (8.131).

A confusing feature for the beginner is the fact that the factorization (F.50)—that is, all terms apart from the triple products of one-body GF's —*does not give the correct lowest-order skeleton term* of (F.48), that is, it does not reproduce expression (F.49). The factorization is *only valid for the correlated parts*, that is, for the second-, third-, fourth-, fifth-, and all corresponding higher-order diagrams (Fig. F.3) contained in (F.48). In order to adjust the correct lowest-order contribution, we have to add to the factorization the triple product of $G(i, j)$'s in (F.50). On the other hand, the correlated parts sum up the infinite sub-series contained in the exact two-body correlation functions figuring in (F.50), therefore factorization may be a powerful approximation.

We recognize that the first term on the rhs of (F.50) sums up the particle–particle (*pp*) correlations (see Fig. F.3) and these are the important ones for the treatment of short-range correlations introduced by the hard core of the bare nucleon–nucleon force, as we discussed in Section 4.3. Therefore, if we want to treat only those correlations, we can drop the other terms in (F.50) and write for the mass operator:

$$M(1,1') = T(1,2;1',4)G(4,2) \tag{F.52}$$

with the T-matrix

$$T(1,2;1',4) = \bar{v}(1,2;1',4) + \tfrac{1}{4}\bar{v}(1,2;3'4')G(3',4';5,6)\bar{v}(5,6;1'4). \tag{F.53}$$

There exists a well-known integral equation for the two-body GF figuring in (F.53) (see below) [Mi 67]:

$$G(1,2;1',2') = G^0(1,2;1',2') - \tfrac{1}{4}G^0(1,2;3,4)K^{pp}(3,4;3',4')G(3',4';1',2'), \tag{F.54}$$

*The response function defined in (8.130) is in time space for $t_1 = t_1'$ and $t_3 = t_3'$ given by $\tilde{R}(1,3;1'3') = -i\Theta(t_1 - t_3)\langle 0|[a^+(1')a(1), a^+(3')a(3)]|0\rangle$. This definition differs from (F.51) by the sign of $i\eta$ in the second term of (8.130), by an additional factor $(-i)$ and by the order of the fermion operators. Since the physical content of both definitions is the same, we can work with both functions.

where K^{pp} is an effective pp-force and to lowest order is given by

$$K^{pp}(3,4;3',4') \simeq \bar{v}(3,4;3',4'). \tag{F.55}$$

If we furthermore replace the one-body GF's in (F.54) by their HF expression using (F.28, 29) and neglect in (F.28, 29) the second part (hole contribution; it can be shown that the hole lines are of higher order in the density and are therefore supposed to be smaller than the particle contribution; see, e.g., [Ma 67b]), we obtain with (F.52–55) the structure of the Brückner–Hartree–Fock equation (5.72). We now have to recognize that the T-matrix (F.53) depends effectively only on one time difference, giving rise to an energy dependence after Fourier transformation. If this energy dependence is replaced by the prescription (5.75), we can identify Eqs. (F.52) and (F.53) exactly with the Brückner HF equations (5.72).

F.5 Hartree–Fock–Bogoliubov Equations

It is very instructive to derive not only Brückner Hartree–Fock theory but also Hartree–Fock–Bogoliubov theory using GF techniques. For this purpose we again retain only the first term on the rhs of (F.50), because we already know that pairing correlations are pp-correlations. The hole-GF $G(1, 1')$ in the first term of (F.50) is calculated using only the static part of the mass operator. For the two-body GF we need to give the spectral representation:

$$\begin{aligned} G_{12,1'2'}(t-t') &= (-i)^2 \langle 0|T\{a_1(t)a_2(t)a_{2'}^+(t')a_{1'}^+(t')\}|0\rangle \\ &= (-i)^2 \Big\{ \Theta(t-t') \sum_{\rho^+} \langle 0|a_1 a_2|\rho^+\rangle\langle\rho^+|a_{2'}^+ a_{1'}^+|0\rangle e^{-(i/\hbar)(E_\rho^{A+2}-E_0^A)(t-t')} \\ &\quad + \Theta(t'-t) \sum_{\rho^-} \langle 0|a_{2'}^+ a_{1'}^+|\rho^-\rangle\langle\rho^-|a_1 a_2|0\rangle e^{-(i/\hbar)(E_0^A - E_\rho^{A-2})(t-t')} \Big\} \end{aligned} \tag{F.56}$$

or, after Fourier transformation,

$$G_{121'2'}^{(\omega)} = (-i)\left\{ \sum_{\rho^+} \frac{\langle 0|a_1 a_2|\rho^+\rangle\langle\rho^+|a_{2'}^+ a_{1'}^+|0\rangle}{\omega - \Omega_\rho^{pp} + i\eta} - \sum_{\rho^-} \frac{\langle 0|a_{2'}^+ a_{1'}^+|\rho^-\rangle\langle\rho^-|a_1 a_2|0\rangle}{\omega - \Omega_\rho^{hh} - i\eta} \right\},$$

$$\hbar\Omega_\rho^{pp} = E_\rho^{A+2} - E_0^A, \qquad \hbar\Omega_\rho^{hh} = E_0^A - E_\rho^{A-2}, \tag{F.57}$$

where the $E_\rho^{A\pm 2}, E_0^A$ are exact eigenenergies.

The HFB theory implicitly assumes that in the ρ-sum of (F.57) only the ground states are relevant, that is, they are supposed to be so collective that in the sum of the residues of (F.57) (which is a constant) the residue of the ground state gathers almost all the strength. As we explained in Chapter 11, the notion of the collectivity of a specific mode is an essential feature of a phase transition—here, the transition to a superfluid state. Remembering that the single-particle GF of the first term in (F.50) becomes a hole line in the mass operator, we get [using (F.21), (F.22), and

(F.50)] the following approximate Dyson equation, in which we have isolated the so-called Cooper poles [Pi 61] in the dynamic part of the mass operator* [for the derivation, one should work entirely in time–space using (F.29) and Fourier transform only at the end]:

$$(\hbar\omega-\epsilon_1)G_{11'}(\omega)=\delta_{11'}+\bar{v}_{1234}\rho_{42}G_{31'}(\omega)+M_{12}^{\text{Cooper}}\cdot G_{21'}(\omega),$$

$$M_{11'}^{\text{Cooper}}(\omega)=\Delta_{12}^{(+)}\cdot\left(\frac{\rho}{\hbar\omega-\hbar\Omega_0^{pp}+h^*}\right)_{22'}\cdot\Delta_{2'1'}^{(+)*}+\Delta_{12}^{(-)}\cdot\left(\frac{1-\rho}{\hbar\omega-\hbar\Omega_0^{hh}+h^*}\right)_{22'}\cdot\Delta_{2'1'}^{(-)*},\tag{F.58}$$

with

$$\begin{aligned}\Delta_{12}^{(+)}&=-\tfrac{1}{2}\bar{v}_{1234}\langle 0|a_3a_4|A+2,0\rangle,\\ \Delta_{12}^{(-)}&=-\tfrac{1}{2}\bar{v}_{1234}\langle A-2,0|a_3a_4|0\rangle.\end{aligned}\tag{F.59}$$

We can forget about the infinitesimal $\pm i\eta$ in the denominators of (F.58) because we shall only consider bound state problems. The next approximation we can make for the case of a phase transition to a superfluid state is:

$$\begin{aligned}\hbar\Omega_0^{pp}&\simeq 2\left(E_0^{A+1}-E_0^{A}\right)=2\lambda^{(+)},\\ \hbar\Omega_0^{hh}&\simeq 2\left(E_0^{A}-E_0^{A-1}\right)=2\lambda^{(-)}.\end{aligned}\tag{F.60}$$

This approximation implies that the correlation energy of the two particles forming the Cooper pair is zero. This, of course, is another assumption we can make in treating a phase transition: the specific mode becomes so soft that its excitation energy goes to zero. We finally make a further (number nonconserving) approximation in assuming that for large systems (this approximation can also be avoided [Mi 67, Sec. 1.4.6]):

$$\Delta^{(+)}\simeq\Delta^{(-)}=\Delta;\qquad \lambda^{(+)}\simeq\lambda^{(-)}=\lambda.\tag{F.61}$$

Thus we arrive at the so-called *Gorkov equations* [Go 58]:

$$\begin{aligned}(\hbar\omega'-h')_{12}G_{21'}&=\hbar\delta_{11'}-\Delta_{12}F_{21'}^*,\\ (\hbar\omega'+h'^*)_{12}F_{21'}^*&=\Delta_{12}^*G_{21'},\end{aligned}$$

or

$$\begin{pmatrix}\hbar\omega' & 0\\ 0 & \hbar\omega'\end{pmatrix}_{12}\begin{pmatrix}G\\ -F^*\end{pmatrix}_{21'}=\begin{pmatrix}\hbar\delta_{11'}\\ 0\end{pmatrix}+\begin{pmatrix}h' & \Delta\\ -\Delta^* & -h'^*\end{pmatrix}_{12}\begin{pmatrix}G\\ -F^*\end{pmatrix}_{21'},\tag{F.62}$$

where we have made the identification $\hbar\omega'=\hbar\omega-\lambda$, $h'=h-\lambda$, and

$$F_{11'}^*=-\left(\frac{1}{\hbar\omega'+h'^*}\right)_{12}\Delta_{22'}^*G_{2'1'}.\tag{F.63}$$

* We should notice that for a hole line the direction of the Dyson equation is inverted and, therefore, the complex conjugate of the single-particle potential is figuring in (F.58). It can also be verified perturbatively for the R-function in (F.39). This, however, becomes important only for time-reversal non-invariant single-particle Hamiltonians.

On the rhs of (F.62) we recognize the familiar HFB matrix $\mathcal{H}$ of Eq. (7.39). As usual, we can write down a spectral representation for $G(1, 1')$ (F.6a, b) and from (F.63) we see that this is also possible for $F^*(1, 1')$:

$$\begin{aligned} G_{11'}(\omega) &= \hbar \sum_k \frac{V_{1k}^* V_{1'k}}{\hbar\omega + E_k - i\eta} + \frac{U_{1k} U_{1'k}^*}{\hbar\omega - E_k + i\eta}, \\ -F_{11'}^*(\omega) &= \hbar \sum_k \frac{U_{1k}^* V_{1'k}}{\hbar\omega + E_k - i\eta} + \frac{V_{1k} U_{1'k}^*}{\hbar\omega - E_k + i\eta}. \end{aligned} \tag{F.64}$$

From (F.64) we find that at the poles Eqs. (F.62) are identical to the HFB equations (7.42).

In order to close the system of equations we have to show how Δ is obtained from the solutions $G_{11'}, F_{11'}^*$. For this purpose we write for the amplitudes appearing in (F.59):

$$\begin{aligned} &\langle 0|a_3 a_4|A+2, 0\rangle \\ &= \lim_{\substack{t=t_2=t_1+0\to-\infty \\ t_3=t_4+0\to 0}} \exp(-i\Omega_0^{pp} t) \frac{\langle 0|T\{a(3)a(4)a^+(2)a^+(1)\}|0\rangle}{\langle A+2, 0|a_2^+ a_1^+|0\rangle}. \end{aligned} \tag{F.65}$$

This relation [WE 70] can be shown to be true in passing from the time limit to the Abelian limit:

$$\begin{aligned} |A+2, 0\rangle &= \lim_{\eta\to 0} \eta \int_{-\infty}^{0} dt\, e^{\eta t} \frac{e^{-i\Omega_0^{pp} t}}{\langle A+2, 0|a_2^+ a_1^+|0\rangle} a_2^+(t) a_1^+(t)|0\rangle \\ &= \lim_{\eta\to 0} \frac{\eta}{\langle A+2, 0|a_2^+ a_1^+|0\rangle} \sum_\tau \frac{\langle A+2, \tau|a_2^+ a_1^+|0\rangle}{-i\left(E_0^{A+2} - E_\tau^{A+2} + i\eta\right)} |A+2, \tau\rangle \\ &= |A+2, 0\rangle. \qquad \text{Q.E.D.} \end{aligned} \tag{F.66}$$

For the two-body GF appearing in (F.65), we can make use of the integral equation (F.54); replacing one of the one-body GF's in front of K^{pp} by its lowest-order expression (F.26) and inserting (F.54) with (F.55) into (F.65), we obtain, with (F.61) and (F.63), the "gap equation":

$$\Delta_{12} = \tfrac{1}{2} \bar{v}_{1234} \frac{1}{2\pi i} \oint d\omega\, F_{34}(\omega). \tag{F.67}$$

Therefore, together with (F.62) we have a closed system of equations. The fact that we have replaced one of the one-body GF's in (F.54) by its lowest-order expression (F.26) may seem a somewhat asymmetric treatment of the two one-body GF's in (F.54); it is, however, consistent with the approximation (F.58) of the mass operator, as the derivation of the Gorkov equations (F.62) has shown.

We would like to make a graphical and physical interpretation of the set

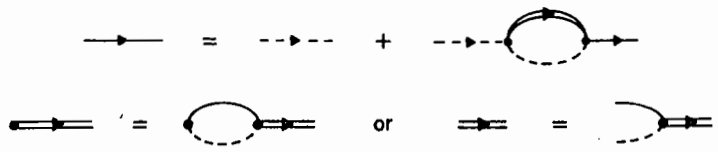


Figure F.4 (a) Graphical representation of (F.58), where the broken lines represent the lowest-order approximation (F.26) and ⇉ stands for the pairing mode. (b) Graphical representation of the gap equation (F.67).

of coupled equations we have just derived. Equation (F.58) is graphically represented in Fig. F.4a and Eq. (F.67) in Fig. F.4b.

From Fig. F.4 we see that the content of the HFB equations in the language of GF's is that the pairing mode is calculated with single-particle propagators which themselves are renormalized by the coupling of a particle to the pairing mode and a hole. The solution of these coupled equations implicitly sums up an important part of the perturbation series, thus allowing the description of a phase transition.

The fact that for the derivation of Brückner HF theory and HFB theory we initially used the same approximation for the mass operator [first term on the rhs of (F.50)] may be perplexing. It should be recognized, however, that Brückner HF theory never isolates the Cooper pole contained in the two-body GF of (F.50). The fact that we treat this pole explicitly in the HFB theory is therefore not in conflict with Brückner HF, and we can thus also combine both theories within the Brückner HFB theory.

Until now we have treated phase transitions from a normal to a superfluid state, and we have shown how the fact that one particular *pp*-mode takes up almost all collectivity, and as a consequence becomes very soft, allows us to derive the HFB equations. From our earlier considerations (see Chap. 7), we know that the HFB equations allow for spherical (rotational symmetry conserving) and deformed (rotational symmetry breaking) solutions. In our treatment of the pairing phenomenon, however, we have nowhere assumed that a breaking of rotational symmetry can take place—we have always supposed that $\rho_{11'}=\langle 0|a_{1'}^{+}a_1|0\rangle$ is a density corresponding to a ground state having good angular momentum (i.e., $I=0$ for the even–even nuclei to fix ideas). From what was stated in Chapters 7 and 11 about the phase transition from a spherical to a deformed nucleus, we know that this is just associated with another soft mode—the quadrupole vibration. We will show in more detail how this can be described in the GF formalism.

It is obvious that in order to deal with the transition from a spherical to a deformed state we have to isolate the coupling of a particle to the quadrupole phonon in the mass operator (F.25). This can be most easily achieved by retaining from the factorization (F.50) only those terms in which the response functions (F.51) appear. The spectral decomposition then yields the excitation energies of the A system (8.130) (see footnote

page 633):

$$R_{13,1'3'}(\omega)=i\sum_{\mu>0}\frac{\langle 0|a_{1'}^{+}a_3|\mu\rangle\langle\mu|a_{3'}^{+}a_1|0\rangle}{\omega-\Omega_\mu^{ph}+i\eta}-\frac{\langle 0|a_{3'}^{+}a_1|\mu\rangle\langle\mu|a_{1'}^{+}a_3|0\rangle}{\omega+\Omega_\mu^{ph}-i\eta}. \tag{F.68}$$

As in the case of the pairing mode, we now assume that the lowest of the excitations in (F.68) practically exhausts all the strength, and that its excitation energy goes to zero ($\Omega_\mu^{ph}\to 0$). Here we assume that this is usually the case for the quadrupole mode (indeed, its excitation energy becomes very low for rotational nuclei), however, it is sometimes possible (^{208}Pb) for other modes to become lowest in excitation. For the single-particle GF's figuring in (F.50) we take again the lowest-order approximation (F.26). The Dyson equation thus takes the following form.

$$(\hbar\omega-\epsilon_1)G_{11'}(\omega)=\delta_{11'}+\bar{v}_{1234}\rho_{42}G_{31'}(\omega)+M_{12}^{\text{phonon}}G_{21'}(\omega), \tag{F.69}$$

with

$$M_{11'}^{\text{phonon}}(\omega)=Q_1^{\sqrt{2+2}}\left(\frac{1}{\hbar\omega-h}\right)_{22'}Q_{1'}^{\sqrt{2+2'*}} \tag{F.70}$$

and

$$Q_1^{\sqrt{2+2}}=\bar{v}_{1(324)_{j_1}}\langle 0|\left[a_3^{+}\,a_4\right]_{2^+}|2^+\rangle. \tag{F.71}$$

It should be noted that until now we have not made any symmetry-violating approximation, since—as has been indicated—everything is coupled to good angular momentum (we couple the particle 2 or 2′ to the quadrupole phonon to give the angular momentum j_1 of the external particle). We should also recognize that the Q-field of (F.71) corresponds to the γ-vertices of particle vibration coupling theory in Section 9.3.3 (with no factor $\frac{1}{2}$; see discussion footnote page 386). We can again write (F.69, 70) in the form of two coupled equations by introducing

$$P_{11'}=-\left(\frac{1}{\hbar\omega-h}\right)_{12}Q_{2'}^{\sqrt{2+2*}}G_{2'1'}, \tag{F.72}$$

and therefore obtain

$$\begin{aligned}(\hbar\omega-h)_{12}G_{21'}(\omega)&=\delta_{11'}+Q_1^{\sqrt{2+2}}P_{21'}^{*}(\omega),\\ (\hbar\omega-h)_{12}P_{21'}^{*}(\omega)&=-Q_2^{\sqrt{2+1}*}G_{21'}(\omega).\end{aligned} \tag{F.73}$$

As in the pairing case, we have to close the system of equations. This can be done analogously here by using instead of (F.50) the Bethe–Salpeter equation for the response function (8.131). The result is

$$\langle 0|\left[a_3^{+}\,a_4\right]_{2^+}|2^+\rangle=\delta\rho_{43}^{(2^+)}=\frac{1}{2\pi i}\oint d\omega\,P_{43}(\omega). \tag{F.74}$$

Insertion of (F.74) into (F.71) constitutes, together with (F.73), the "gap equation" for the "order parameter" Q of the deformation field.

Two single-particle fields enter into Eqs. (F.76), one involving the ordinary density $\rho_{12}=\langle 0|a_2^{+}a_1|0\rangle$ and the other the transition density

$$\delta\rho_{12}^{(2^+)}=\langle 0|\left[a_2^{+}\,a_1\right]_{2^+}|2^+\rangle.$$

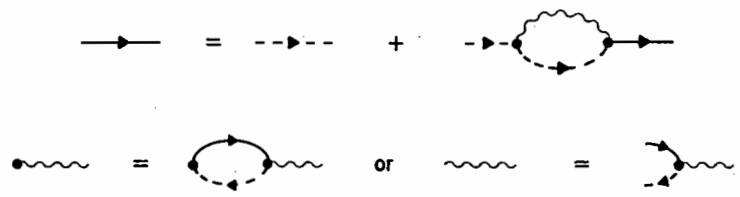


Figure F.5 (a) Graphical representation of Eq. (F.69). The wavy line represents the soft quadrupole phonon. (b) Graphical representation of the "gap equation" for the deformation field Q.

The transition density, of course, projects out the quadrupole part of the *ph* force in (F.71) and the ordinary density projects out the monopole part of the force in (F.69). We again wish to stress the point that no symmetry-violating approximation has been made so far, as we have coupled everything to good angular momentum. The symmetry breaking of deformed HF theory can be introduced by saying that the 2^+ becomes identical with the ground state of the internal system. At this moment, having angular momentum coupling no longer makes sense and Eqs. (F.73) just reduce to the usual deformed HF equations. We explicitly find two potentials V_0 and Γ_2, introduced in Sec. 11.2, which now constitute an angular momentum nonconserving single-particle Hamiltonian of the internal system.

As we have already stated, the "gap equation" for the deformation field (F.71, 74) can again be derived from the Bethe–Salpeter equation for the response function (8.131). From (F.74) and (F.72) we see, however, that it can be represented graphically as shown in Fig. F.5b. Eq. (F.69) is represented in Fig. F.5a. Again, the typical nonlinear coupling of single-particle motion and collective vibration becomes evident from Fig. F.5a, b.

In general, we should, of course, separate the Cooper pole and the phonon poles of all multipolarities in the mass operator simultaneously and thus generate the general single-particle theory we treated in Chapter 7.

From (F.71) and (F.59), we see that the transition densities enter the deformation and pair fields. As we have seen that these are peaked at the nuclear surface (Chaps. 8 and 13), it may be a good approximation to replace the forces entering (F.71) and (F.59) by separable ones. This leads naturally to the *pairing-plus-quadrupole model* discussed in Chapter 7, and thus we again see the meaning of this model within the context of nuclear phase transitions.

From the GF treatment of this theory, it also becomes clear how a possible generalization might work: the essential feature would be to solve the RPA modes *self consistently* with the single-particle states, that is, where the single states are coupled back to the RPA modes in the way shown in Fig. F.5a, b. Also, the symmetry-conserving formulation (F.62) of HFB should be an interesting variant of the theory; in fact a much more elaborate theory of this kind has been proposed by Janssen [Ja 79].

F.6 The Bethe–Salpeter Equation and Effective Forces

In the course of the derivation of the HFB equations we have made reference to Bethe–Salpeter equations (BSE) for two-body GF's on several occasions. Since these equations are derived in many textbooks [Mi 67, AGD 65, FW 71], we shall be very brief here. The BSE for the response function (F.51) is given by:

$$R(1,3;1',3') = G(1,1')G(3,3') - G(3,5)G(6,1')K^{ph}(5,7;6,8)R(1,8;7,3'). \quad \text{(F.75)}$$

Equation (F.75) is an exact equation, where K^{ph} is called an effective particle–hole force; it sums up skeleton graphs of the response function, which cannot be cut into two parts by cutting only two lines. The BSE (F.75) is graphically represented in Fig. F.6, together with a typical second-order contribution to K^{ph}. We see that these contributions to K^{ph} can be cut into two halves only by cutting four lines. Equation (F.75) can be verified to any order skeleton perturbation. Repeated indices in (F.75) indicate, as usual, summation over quantum numbers and integration over times. It should be mentioned that (F.75) is sometimes convenient in deriving formal relations. Going beyond the RPA treatment (F.75), however, is extremely complicated, because the kernel generally depends on three times (after taking into account translational invariance with respect to time) or three energies. For practical purposes we sometimes, therefore, use an integral equation for the response function with a kernel depending only on one energy [Sch 71c, Sch 76]. In this way, we arrive at a Dyson equation for bosons (RPA) with a boson mass operator which is quite analogous to the Dyson equation (F.22) for fermions [SE 73, Sch 76]. This treatment of correlation functions is similar in spirit to what has now become known as Mori theory [Mo 65] in the field of condensed matter. To nuclear physics this has been applied by Werner [We 76, 78].

Everything we have said about the BSE for the response function also holds good for the analogous BSE of the two-body GF (F.54):

$$G(1,2;1'2') = G^0(1,2;1',2') - \tfrac{1}{4}G^0(1,2;3,4)K^{pp}(3,4;3',4')G(3',4';1',2').$$

Here K^{pp} again sums up all graphs contributing to $G(1,2;1',2')$ which cannot be cut in two by just cutting two skeleton lines. In Fig. F.7 we give a graphical representation of (F.54) together with a second-order skeleton

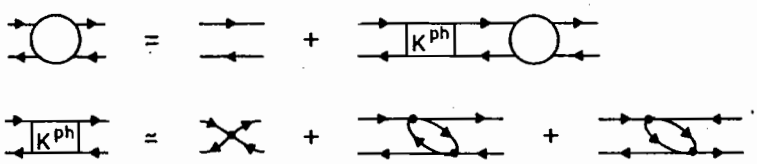


Figure F.6. Bethe-Salpeter equation for the response function (F.50) and schematic representation of lowest-order contributions to K^{ph}.

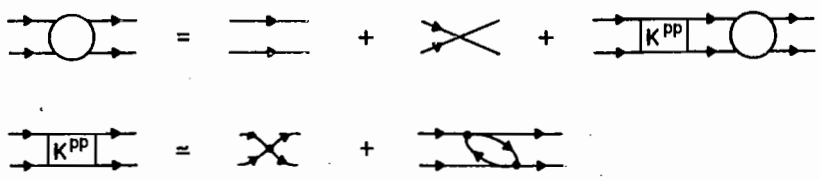


Figure F.7. Bethe-Salpeter equation for the *pp* GF of (F.76) and schematic representation of the lowest order contribution to K^{pp}.

contribution to K^{pp}; the lowest order being, of course, the bare interaction itself.

As we have shown above, the gap equations for the pair field Δ (F.67) and for the deformation field Q (F.71) can be derived from the corresponding BS equations (F.54) and (F.75). It is therefore obvious that these order parameters have to be calculated with the corresponding effective *pp* and *ph* forces K^{pp} and K^{ph}, respectively. To show this explicitly is somewhat tricky, and presenting details would be very clumsy, so we will only sketch the main features here.

We have already shown that in the gap equations for Δ and Q we have to use K^{pp} and K^{ph}, respectively. It is more difficult to show how K^{pp} and K^{ph} come into play in the first of equations (F.62) and (F.73), respectively. Let us denote the R-function appearing in the mass operator (F.25) by R_6; then for the mass operator we can write in an obvious shorthand notation

$$M^r(\omega) = \bar{v} R_6 R_6^{-1} R_6 \bar{v}. \tag{F.76}$$

We have seen that HFB theory corresponds to approximating R_6 by the first term on the rhs of (F.50), which we will denote by GG_4. In this approximation we also have, of course,

$$R_6^{-1} = [GG_4]^{-1}. \tag{F.77}$$

We now observe that from (F.45) and (F.46) we have

$$G(1,2;1',2') = G^0(1,2;1',2') - \frac{i}{2} R(1,2,3^{++};1'',2',3'^{+}) \bar{v}(3',1'';3,1''') \cdot G(1'',1'). \tag{F.78}$$

Together with (F.54), we deduce the identity:

$$R(1,2,3^{++};1'',2',3'^{+}) \bar{v}(3',1'';3,1''') = G(2,3^{++};2',3'^{+}) \cdot G(1,1'') K^{pp}(3',1'';3,1'''). \tag{F.79}$$

Replacing the parts $R_6\bar{v}$ and $\bar{v}R_6$ in (F.76) by (F.79) and its adjoint, respectively, and using (F.77), we see that in (F.58) and (F.62) we have to use K^{pp} everywhere. Analogously, we can, of course, show that we have to

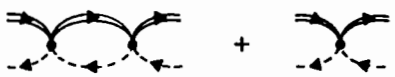


Figure F.8. Hole-pairing mode scattering process omitted in (F.77).

use K^{ph} in (F.69) and (F.73). It can be shown [ES 69] that the terms we have neglected by the approximation (F.77) are of the type shown in Fig. F.8, where the hole scatters from the pairing mode.

This shall conclude our brief outline of GF theory, and for more details of the standard techniques we again refer the reader to the literature cited in this appendix.